

itsmyplan — Master Systems & Prompts Architecture

COMPLETE CAD APPLICATION ENGINE SPECIFICATION FOR ANTIGRAVITY AI ENGINE

1. System Vision & Architecture Strategy

itsmyplan is a professional-grade room renovation, layout planning, and interior detailing system designed to operate completely within user-controlled environments. Rooted fundamentally in the *Scalebanana Principle*, the app focuses on executing a completely stable, ultra-low-latency 2D structural drafting data model, which seamlessly drives a rich 3D visualization canvas layer.

The system explicitly executes under a **Zero Server Framework Architecture**. There are no heavy server infrastructures, data collectors, middleware tracking loops, or unoptimized data structures. The runtime stack relies on Next.js client-side execution, React 19 hooks for instantaneous mechanical mathematics, Tailwind CSS for strict UI density, and native layout elements (SVG for 2D, WebGL/Three.js for 3D) to maximize rendering speed.

2. Technical Domain Rules & Mechanics

- **Absolute Metric Unit Uniformity:** To guarantee accuracy for architectural components, structural framing, and plumbing, all dimension figures are strictly parsed, stored, and mutated in integers representing millimeters (`mm`). Unit conversion happens only at rendering boundaries.
- **Metric Snapping Engine:** Mouse coordinate streams canvas-wide clamp automatically to a sharp, deterministic `50mm` interval grid matrix, preventing minor dimensional offsets during wall composition or asset alignment.
- **Dual Surface Pipeline:** The UI maintains synchronized data rendering pipelines. The 2D component handles vector tracing, spatial clearance evaluations, and material takeoff lookups via an explicit SVG scene tree. Concurrently, a passive 3D frame context interprets layout state mutations to generate instant real-time dollhouse perspectives via React Three Fiber.

3. Complete UK Structural & Furniture Asset Matrix

The asset layer is embedded with real-world metric dimensions optimized for structural mapping, space clearing, and interior overhaul pipelines:

CLASSIFICATION	ASSET TYPE IDENTIFIER	STANDARD METRICS (W × D)	CLEARANCE / ACTIVITY ZONES
Bathroom	uk-bath-standard	1700 mm × 700 mm	700mm clearance path along longitudinal edge
Bathroom	uk-bath-compact	1500 mm × 700 mm	700mm clearance path along longitudinal edge
Bathroom	uk-wc-close-coupled	400 mm × 650 mm	700mm × 700mm clear frontal footprint box
Bathroom	uk-wc-wall-hung	380 mm × 550 mm	700mm × 700mm clear frontal footprint box
Bathroom	uk-basin-pedestal	550 mm × 450 mm	700mm × 700mm operational workspace arc
Bathroom	uk-basin-cloakroom	450 mm × 250 mm	600mm × 600mm front clearance footprint
Bathroom	uk-shower-square	900 mm × 900 mm	700mm clear access zone outside the enclosure door
Bathroom	uk-shower-quadrant	900 mm × 900 mm	700mm clear entry path along curved dynamic sweep
Kitchen	uk-kitchen-base-standard	600 mm × 600 mm	600mm frontal operational zone for drawer extraction
Kitchen	uk-kitchen-base-slim	400 mm × 600 mm	600mm frontage clearance pathway line
Kitchen	uk-kitchen-appliance-tower	600 mm × 600 mm	750mm clearance threshold to handle hot door utility
Kitchen	uk-kitchen-sink-double	1000 mm × 500 mm	700mm continuous functional standing clearance lane
Bedroom	uk-bed-double	1350 mm × 1900 mm	600mm continuous walking pathway around three sides
Bedroom	uk-bed-king	1500 mm × 2000 mm	650mm walking perimeter access lane clearances
Bedroom	uk-wardrobe-standard	600 mm × 600 mm	600mm structural swinging door operational tolerance
Bedroom	uk-wardrobe-slider	1200 mm × 500 mm	500mm standard sliding corridor access clearance

4. Core Application Data Schemas

The system data state must maintain strict static structures, guaranteeing that export files, database payloads, and computational loop layers map identically across execution domains.

```
export interface Point {
  x: number; // Coordinate placement along X plane in mm
  y: number; // Coordinate placement along Y plane in mm
}

export interface Wall {
  id: string;
  p1: Point; // Vector starting point position node
  p2: Point; // Vector terminating point position node
  thickness: number; // Default: 100mm (internal partition), 300mm (cavity exterior)
}

export interface Opening {
  id: string;
  wallId: string; // Relational mapping boundary linking the opening to a Wall
  type: 'door' | 'window';
  distanceFromP1: number; // Distance interval in mm shifting from wall node p1 toward p2
  width: number; // Horizontal metric structural sizing span in mm
  height: number; // Vertical structural span in mm (essential for takeoff subtractions)
  zOffset: number; // Baseline vertical translation clearance from the subfloor level in mm
  flipDirection: boolean; // Structural vector axis modifier (swings/orientations)
}

export interface Fixture {
  id: string;
  type: string; // Maps directly to standard library codes (e.g., 'uk-bath-standard')
  x: number; // Plan view horizontal midpoint vector position in mm
  y: number; // Plan view vertical midpoint vector position in mm
  rotation: number; // Angular orientation constraints locked to: 0, 90, 180, 270 degrees
  showClearance: boolean; // Dynamic tracking property visibility toggle
}

export interface AppViewSettings {
  showDimensions: boolean;
  showClearanceZones: boolean;
  showServiceLayers: boolean;
  activeMode: '2d' | '3d' | 'takeoff';
}
```

5. Multi-Page Print Layout Specification

The PDF engine compiles high-density layout summaries purely within client execution space without transferring payload packages to an external print server or processing microservice.

1. **Page 1 — High-Fidelity 2D Scalable Layout:** The print engine iterates through active structural canvas configurations, translating nodes directly into precise path metrics within a native client compilation document framework (e.g., j sPDF). Text fields write out precise vector metrics alongside a formal architectural title block

including project parameters, design dates, scales, and user references. All lines are outputted as true vector drawing operations to prevent scaling pixelation during reproduction.

2. **Page 2 — High-Resolution 3D Presentation View:** To extract the 3D perspective context, the application enforces a continuous frame buffer presence via WebGL settings (`preserveDrawingBuffer: true`). The application pulls a compressed base64 image data payload straight from the WebGL context buffer surface matching actual pixel densities, embedding the target raster view beautifully centered within structural report block borders.

6. The Antigravity System Master Prompt (v3 Production Bundle)

Provide the comprehensive instruction block below into your agentic AI engineering console to build or update the core architecture of your application:

MASTER PROMPT INGESTION FRAME — COPY INTACT

Role & Context

You are a Principal Software Architect specializing in lightweight interactive CAD layout frameworks, computational math pipelines, and high-performance, client-side rendering engines.

System Objective

Build "itsmyplan" — a high-performance, browser-isolated 2D Room Layout Planner featuring an instant real-time 3D Dollhouse Preview panel, fully integrated Firebase Authentication paths with offline preservation fallbacks, smart clearance validation loops, material takeoff reporting calculators, and comprehensive multi-page PDF blueprint print capabilities.

Core Philosophy (Scalebanana & Zero Server)

Adhere strictly to the Scalebanana Principle: the foundational application structure, computational grid geometries, calculation state listeners, and document generation workflows must establish perfect mathematical reliability before structural cosmetics are introduced. Operate entirely under a Zero Server structure: all logic, data parsing, rendering contexts, and export processing must run entirely in the browser.

Technical Framework Stack

- Environment Platform: Next.js (App Router, completely optimized for Client-Side Rendering)
- Core Reactive Engine: React 19 & Clean Component State Custom Hooks
- Design / Presentation Architecture: Tailwind CSS
- 2D Precision Tracing Interface: High-Performance Native SVG Elements
- 3D Visualization Pipeline: Three.js executed through React Three Fiber (R3F) and @react-three/drei
- PDF Compilation Engine: Client-side jsPDF or pdfmake orchestration layers
- Persistence Mechanism: Dual-Layer Pipeline (Firebase Auth + Cloud Firestore, falling back seamlessly to LocalStorage for offline/anonymous users)

Essential Global Engineering Constants

- CEILING_HEIGHT = 2400 // Fixed standard vertical ceiling projection wall bound in mm
- SNAP_INTERVAL = 50 // Cursor tracking increments locked cleanly to 50mm vectors

Critical Subsystem Engineering Deliverables

1. Dual-Viewport Workspace Interface

- Build a responsive horizontal side-by-side or tabbed workspace layout using Tailwind CSS. Left frame serves as the interactive 2D drafting floor plan panel; right frame exposes a real-time React Three Fiber WebGL 3D preview layer updating continuously as state variations manifest.

2. High-Precision Vector 2D Snapping Grid

- Construct an active coordinate tracing loop handling canvas input boundaries. Clamp active mouse movements directly to the standard 50mm grid interval. Provide user tracing lines with real-time dynamic length text indicators (e.g., "3650 mm") positioned relative to active points. All walls must connect accurately.

3. Integrated UK Asset Blueprint Library

Provide a dense sidebar array exposing standard pre-configured metric structural component definitions across Bathroom, Kitchen, and Bedroom domains. Assets drop onto the canvas center on click and can rotate by 90-degree steps via the "R" key shortcut:

- Bathroom: Standard Bath (1700x700mm), Compact Bath (1500x700mm), Close-Coupled WC (400x650mm), Wall-Hung WC (380x550mm), Pedestal Basin (550x450mm), Cloakroom Basin (450x250mm), Square Shower (900x900mm), Quadrant Shower (900x900mm).
- Kitchen: Base Unit Standard (600x600mm), Base Unit Slim (400x600mm), Appliance Tower (600x600mm), Double Sink Module (1000x500mm).
- Bedroom: Double Bed (1350x1900mm), King Bed (1500x2000mm), Wardrobe Standard (600x600mm), Wardrobe Slider (1200x500mm).

4. Translucent Ergonomic Clearance Zones

- Map custom clearance activity areas around all library fixtures as defined in the technical specification. Provide a top-level UI toggle switch to render these clearance fields as soft, translucent colored shapes (e.g., `` or `` with Tailwind alpha channels) surrounding the active object boundary.

5. Automated Structural Material Takeoff Engine

- Implement a pure JavaScript analytic utility layer computing architectural material dimensions based on current layout states:
 - * Total Floor Area (m²): Compute flat horizontal space contours.
 - * Total Wall Surface Area (m²): Compute aggregate wall line lengths multiplied by the standard 2400mm vertical ceiling height, mathematically subtracting the exact dimensions of integrated doors and windows.
 - * Total Base Perimeter (Linear Metres): Track total baseline metrics for materials estimation like skirting board runs.
- Expose these parameters within a dedicated Tailwind overlay reporting container called "Material Takeoff Sheet".

6. Firebase Identity Routing & Local Portability Engine

- Setup standard Firebase Auth integration interfaces. If authenticated, stream user blueprints straight to Firestore under isolated document records: `/users/{uid}/plans/{planId}`. Ensure no administrative global cross-reads occur. If working offline or unauthenticated, fall back seamlessly to LocalStorage keys.
- Build independent file data transfer actions:
 - * Export Project: Assemble active structural state objects into a structured JSON payload wrapper and prompt an instant client-side file browser download (`.json`).
 - * Import Project: Read ingested file structures using the browser's FileReader API, evaluate validity parameters, and instantly restore the structural canvas context.

7. Dual-Layer Multi-Page PDF Document Exporter

- Build a single client execution worker parsing layout states into a clean document structure:
 - * Page 1 (2D Layout Blueprint): Map SVG elements directly to sharp, native PDF lines using exact metric scale calculations. Append a formal project parameters title block layout.
 - * Page 2 (3D Render Frame): Build the R3F canvas context with `preserveDrawingBuffer: true`. Trigger a canvas buffer snapshot sequence via `gl.canvas.toDataURL()`, appending the uncompressed rendering frame centered cleanly inside a professional layout report framework.

8. Web Standards Compliance Integration

- Include a structural layout placeholder module mapping essential site text assets and legal structures. Name this specific operational framework bundle "COMPLIANCE MD FOR FUTURE APPS". It must provide modular, accessible layout references for a Cookie Selection Banner, Terms of Service agreements, privacy policies, cookie disclosure sheets, and an Accessibility Statement.

9. Structural Isolation Mandate

- Keep canvas logic loops, mouse track clamping operations, and rendering conversions fully isolated within a dedicated React Custom Hook (`useRoomEditor``). Do not use remote server endpoints, tracking cookies, or analytical event logging tools. Deliver clean, highly maintainable code blocks using explicit TypeScript syntax typing loops throughout.